

Case Study: Optimizing Global E-Commerce Performance



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WORLD ANALYSTS & CONSULTANTS AGENCY

Role: Data Analytics Consultant

Trusted Analytics • Smarter Decisions

Tools: SQL (BigQuery/MySQL), Power BI



Sales & Revenue Analysis



Business Intelligence
Dashboards

Dataset: Amazon Global Sales Performance



Operational & Inventory
Analytics



Strategic Data Consulting

1. Executive Summary



Executive Reporting & Insights



Executive Reporting & Insights

This project analyzes a global Amazon sales dataset to identify key revenue drivers, regional performance disparities, and payment preferences. By leveraging advanced SQL querying and interactive data visualization, I transformed raw transactional data into actionable insights aimed at improving regional marketing ROI and optimizing inventory distribution.

2. Business Challenges

To drive growth, the following strategic questions were addressed:

- **Revenue Concentration:** Which product categories anchor the business?
- **Temporal Trends:** What does the revenue trajectory look like, and are there seasonal risks?

Transforming Data Into Strategic Growth

- **Customer Sentiment:** Does higher customer satisfaction directly correlate with higher spend?

- **Operational Efficiency:** Which payment methods dominate, and are there regional "outlier" transactions requiring investigation?

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3. The Technical Solution (SQL Deep Dive)



Sales & Revenue Analysis



Business Intelligence
Dashboards



KPI & Performance Tracking



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Operational & Inventory
Analytics



Executive Reporting & Insights

I utilized Window Functions, CTEs, and Subqueries to extract deep insights.

Below are the key analytical modules:

A. Regional & Category Performance

Objective: Identify the "Hero" products for each region using Rank functions.

SQL

SELECT

customer_region,

product_category,

SUM(total_revenue) **AS** revenue,

RANK() **OVER** (**PARTITION BY** customer_region **ORDER BY**

SUM(total_revenue) **DESC**) **AS** category_rank

FROM `amazon\_sales\_dataset`

GROUP BY 1,2; — *Transforming Data Into Strategic Growth*

B. Anomaly Detection (Outlier Analysis)

Objective: Identify transactions exceeding 2 standard deviations from the mean to flag potential wholesale orders or data errors.

SQL

SELECT *

FROM `amazon\_sales\_dataset`

WHERE total_revenue > (

SELECT AVG(total_revenue) + 2 * STDDEV(total_revenue)

FROM `amazon\_sales\_dataset`

);

C. Cumulative Growth Trajectory

Objective: Calculate a running total to visualize business scaling over the fiscal year.

SQL

SELECT

order_date,

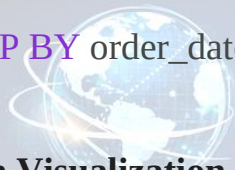
SUM(total_revenue) AS daily_revenue,

SUM(SUM(total_revenue)) OVER (ORDER BY order_date) AS

cumulative_revenue


```
FROM `amazon_sales_dataset`
```

```
GROUP BY order_date;
```



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4. Data Visualization & Insights

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The dashboard translates complex SQL outputs into intuitive visual narratives.

Key Visual Findings:



Sales & Revenue Analysis



KPI & Performance Tracking



Operational & Inventory



Business Intelligence
Dashboards



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Executive Reporting & Insights

- **Balanced Category Portfolio:** Revenue is remarkably evenly distributed across Beauty, Books, Fashion, and Electronics (each contributing ~19-21%), suggesting a resilient, diversified product mix.
- **Regional Dominance:** The **Middle East** leads in total revenue and quantity sold, followed closely by North America.
- **Payment Ecosystem:** Credit Cards and Wallets are the preferred methods, though "Cash on Delivery" remains a significant factor (~20%), indicating a need for localized payment strategies in specific regions.
- **Seasonality Warning:** Both revenue and customer ratings show a gradual downward trend from January through February. This suggests a "Post-Holiday Slump" that may require targeted promotional intervention.

5. Strategic Recommendations

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As a consultant, I propose the following based on the data:

1. **Retention Strategy:** Investigate the decline in customer ratings observed toward Q1. A drop in satisfaction often precedes a drop in long-term LTV (Lifetime Value).

2. **Targeted Marketing:** Since **Beauty** is a top-performing category in the Middle East, launch exclusive regional bundles to capture higher margins.

3. **Payment Optimization:** Given the high usage of **UPI and Wallets**, ensure the checkout UX is optimized for mobile-first users to reduce cart abandonment.

4. **Inventory Planning:** Use the cumulative revenue growth patterns to forecast stock requirements 60 days in advance of the observed peak periods in January.

